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Electrically conductive fiber optic slickline for coiled tubing operations

Abstract

A slickline that includes both electrically conductive and fiber optic capacity. The slickline includes a fiber optic thread or bundle of threads that may be surrounded by an electrically conductive member such as split half shells of copper elements. Further, these features may be disposed in a filler matrix so as to provide a cohesiveness the core of the slickline. So, for example, the line may be more effectively utilized in downhole applications such as coiled tubing operations, without undue concern over collapse or pinhole issues emerging in the line.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 17/653,663 filed on Mar. 7, 2022, which is continuation of U.S. patent application Ser. No. 14/917,630 filed on Mar. 9, 2016, which is a national stage entry of International Patent Application No. PCT/US2013/059779, filed Sep. 13, 2013, the entirety of each of which are herein incorporated by reference.

BACKGROUND

(1) Exploring, drilling and completing hydrocarbon and other wells are generally complicated, time consuming and ultimately very expensive endeavors. In recognition of these expenses, added emphasis has been placed on efficiencies associated with well completions and maintenance over the life of the well. Along these lines, added emphasis has been placed on well logging, profiling and monitoring of conditions from the outset of well operations. Whether during interventional applications or at any point throughout the life of a well, detecting and monitoring well conditions has become a more sophisticated and critical part of well operations.

(2) Initial gathering of information relative to well and surrounding formation conditions may be obtained by way of a logging application. That is, equipment at the surface of an oilfield adjacent to the well may be used to deploy a logging tool in the well. Often, an electrically conductive logging cable is used to deliver the logging tool into the well. The logging cable may include a variety of power and communicative lines incorporated into the cable along with a host of isolating and protective polymer layers. As a result, the cable may be of relatively substantial weight, strength, and profile. Thus, the cable may be well suited for delivery of the logging tool to significant well depths without undue concern over the accommodated load of the tool or perhaps the increasing load of the cable itself.

(3) While well suited to accommodate a significant amount of load, the use of such cables as described above means that the equipment positioned at the surface of the oilfield may be fairly substantial in terms of footprint and power requirements therefor. Similarly, the set up and performance cost of running the operation may also be quite significant.

(4) Due to the costs involved, a cabled logging application as described above is often bypassed in favor of a more cost effective “slickline” logging application. That is, rather than employing an electrically powered and/or electrically communicative cable as in the case of a cabled application, a comparatively lower profile line without such capacity may be employed in a conventional slickline application. The slickline is a relatively unitary line used to deliver the logging tool which includes no electrically conductive wiring. Thus, a downhole battery is provided and data obtained

from the logging application may be processed after retrieval of the logging tool. Notably, however, due to the substantially lower profile and comparatively lower weight of the slickline, considerable expense may be saved. For example, expenses may be affected in terms of reduced surface equipment requirements, as well as reductions in expenses associated with set up, powering, and footprint requirements.

(5) It is also known to deploy a fiber optic tether into a wellbore, either directly or within coiled tubing, which enables the acquisition and transmission of real-time logging information to monitoring or analytical equipment at the surface of the oilfield. Indeed, such a fiber optic tether may be utilized in conjunction with a host of more interventional applications beyond logging. For example, a fiber optic tether may be of particular benefit when used in combination with coiled tubing applications. Specifically, the tether may be pumped through the coiled tubing at the outset of operations to provide real-time feedback regarding any subsequent applications that utilize the coiled tubing.

(6) While adept at providing the benefits of a low-profile and lightweight means of data transmission, the fiber optic tether remains devoid of any electrically conductive capacity. That is, in contrast to the above noted wireline cable, the fiber optic tether is devoid of the heavier electrical cabling. While this is advantageous for reasons noted hereinabove, it means that reliance on a dedicated, space consuming, downhole power source is still required. Thus, the power available to monitoring equipment is limited to the constraints of such a power source.

(7) In addition to the lack of a surface supported power supply, the fiber optic tether faces other added challenges. For example, this type of tether consists of one or more optical fibers or fiber optic threads as noted above, which are bundled and run through a protective metal tube. Thus, in addition to accommodating fiber optic threads, the tether inherently includes a fair amount of void space. The result is that the tether may be prone to collapse. Similarly, the slightest of pinholes into the metal tube during a downhole application may directly expose the threads to damaging downhole or application fluids.

(8) The problems associated with power limitations and current construction of fiber optic tether do not lend to merely adding in another line. For example, inserting a separate electrical line into the metal tube might address the lack of surface available power. However, since a dedicated power line would not strategically occupy void space, such would only increase the overall diameter of the tether as well as the amount of void space. Indeed, when considering insulating and other layers of the electrical line, the tether may begin to take on dimensions and weight more characteristic of wireline. Thus, the advantage of utilizing the tether may begin to be lost.

(9) At present, operators are generally left with the option of choosing between a lightweight fiber optic tether without electrically conductive capacity and a wireline cable that may be far heavier than is required for the job. Yet, due to the potential need for surface based power, the more cumbersome wireline is often the only practical conveyance option available.

SUMMARY

(10) A fiber optic slickline is provided for use in a well. The slickline includes a fiber optic thread to provide communications through the well. Additionally, an electrically conductive member is provided about the thread so as to support power transmission through the well.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a cross-sectional view of an embodiment of a fiber optic electrical core of a fiber optic slickline.

(2) FIG. 2A is an overview of an oilfield with a well accommodating coiled tubing that utilizes slickline with the fiber optic electrical core of FIG. 1.

- (3) FIG. 2B is an enlarged view of the slickline in the well, taken from 2-2 of FIG. 2A.
- (4) FIG. 3A is a cross-sectional view of the fiber optic electrical core of FIG. 1 with protective layering assembled thereover.
- (5) FIG. 3B is a cross-sectional view of the completed fiber optic slickline of FIGS. 2A and 2B.
- (6) FIG. 4 is a cross-sectional view of an alternate embodiment of a fiber optic slickline utilizing the fiber optic electrical core of FIG. 1.
- (7) FIG. 5A is a cross-sectional view of an alternate embodiment of fiber optic electrical core of a fiber optic slickline.
- (8) FIG. 5B is a cross-sectional view of a completed fiber optic slickline utilizing the fiber optic electrical core of FIG. 5A.
- (9) FIG. 6A is a cross-sectional view of another embodiment of an electrically conductive fiber optic slickline and core.
- (10) FIG. 6B is a cross-sectional view of yet another embodiment of an electrically conductive fiber optic slickline and core.

DETAILED DESCRIPTION

(11) Embodiments are described with reference to certain tools and applications run in a well over slickline. As used herein, the term “slickline” is meant to refer to an application that is run over a conveyance line that is substantially below about 0.25 inches in overall outer diameter. That is, as opposed to a higher profile or diameter wireline cable, downhole applications detailed herein are run over a relatively more cohesive or solid state low profile or diameter slickline. Additionally, the embodiments detailed herein are described with reference to particular logging applications and tools. However, slickline embodiments detailed herein may be particularly beneficial when combined with more interventional applications such as those performed with coiled tubing. Regardless, embodiments generally employ a fiber optic electrical core that includes fiber optic thread that with electrically conductive member disposed about the core so as to render a practical and effective electrically conductive fiber optic slickline.

(12) Referring specifically now to FIG. 1, an embodiment of a fiber optic electrical core **100** is shown. This core **100** may be incorporated into a fiber optic slickline **200** as alluded to above so as to provide separately dedicated telemetric and powered capacity to the line **200** (see FIG. 2A). This is achieved by use of one or more fiber optic threads **120**, **125** to provide dedicated telemetry whereas electrical power may be conveyed over a conductive member **150** about the threads **120**, **125**. In the embodiment shown, four separate threads or optical fibers **120**, **125** are utilized which are each jacketed by a conventional polymeric buffer **127**. The conductive member **150** on the other hand is provided as generally c-shaped split shell or halves **155**, **157**. For example, in the embodiment shown, electrically conductive copper halves **155**, **157** surround the entirety of the underlying bundle of threads **120**, **125**. Thus, efficient use of space within the core **100** is maximized without the morphology of the member **150** providing any added void space. As a result, the overall profile of the slickline **200** may be kept below about 0.20 inches in outer diameter (see FIGS. 2A and 2B).

(13) Continuing with reference to FIG. 1, void space is also avoided by the use of a filler matrix **140** encompassing the fiber optic threads **120**, **125**. For example, a silicon-based polymer may be utilized to form a cohesive unit among the threads **120**, **125**. With added reference to FIGS. 2A and 2B, this may be advantageous in helping the entirety of the slickline **200**, including the threads **120**, **125**, to behave in a cohesive and unitary manner. That is, this may be desirable during a downhole application, as opposed to allowing the threads **120**, **125** to respond independently of the remainder of the line **200** which may occur in the absence of a filler matrix **140**.

(14) In an alternate embodiment where cohesive responsiveness of the fiber optic threads **120**, **125** is of comparatively lesser concern, the core **100** may avoid use of a filler matrix **140**. Instead, the threads **120**, **125** may be intentionally placed within a welded steel tube in a loose fashion with a sufficiently thick electrically insulating polymer layer **180** thereabout.

(15) In the embodiment of FIG. 1, recognition of the electrically conductive nature of the surrounding member **150** is similarly recognized with the fiber optic bundle being encapsulated by the noted electrically insulating polymer layer **180**. Similarly, the electrically conductive member **150** may also be surrounded by an insulating polymer jacket **175**. Thus, a fully insulated electrical conveyance is achieved. Additionally, in one embodiment, the power conductive member **150** may also serve as a backup for data transmission. That is, in circumstances where the comparatively less robust fiber optics **120**, **125** suffer micro-bends or other damage, the member **150** may be utilized as a mode of data relay between downhole tools **275** and surface equipment **225** (again see FIGS. 2A and 2B).

(16) Continuing now with reference to FIGS. 2A and 2B, overview and enlarged views of an oilfield **201** are depicted with a well **280** accommodating an embodiment of an electrically conductive fiber optic slickline **200**. That is, the slickline **200** incorporates the fiber optic electrical core **100** referenced hereinabove and shown in FIG. 1.

(17) FIG. 2A in particular, depicts a logging application at the oilfield **201** that is run through a deviated well **280** via coiled tubing **215**. With added reference to FIG. 2B, the slickline **200** is disposed within an interior flow passage **217** of the coiled tubing **215**. The slickline **200** is communicatively coupled to a logging tool **275**. Thus, characteristics of the well **280** over various formation layers **290**, **295** may be detected by way of the tool **275**. Once more, the fiber optic nature of the slickline **200** allows for certain detections, such as, but not limited to, temperature, pressure, and/or vibration readings, to be taken directly with fiber optic threads **120**, **125** apart from the noted tool **275** (see FIG. 1). Such temperature, pressure, and/or vibration readings may be obtained utilized distributed temperature sensing (DTS), distributed pressure sensing (DPS), distributed strain sensing (DSS) or distributed vibration sensing (DVS) utilizing the fiber optic threads **120**, **125** as will be appreciated by those skilled in the art and transmitted to the surface equipment **225**. Additionally, to the extent necessary, the tool **275** may be effectively powered without reliance on a downhole power source. That is, as noted above, the electrically conductive nature of the slickline **200** allows power for the tool **275** to be obtained from surface equipment **225**.

(18) In the embodiment shown, the surface equipment **225** includes a control unit **240** that may be coupled to the coiled tubing **215** and slickline **200** by way of a reel **260**. Thus, as indicated, effective powering of the downhole tool **275** may be surface directed as well as the overall coiled tubing application. Additionally, data transmitted between the tool **275** and surface equipment **225** (such as the control unit **240**) along the fiber optic threads **120**, **125** and/or the conductive member **150** of the slickline **200** may be analyzed in real-time.

(19) In this example embodiment, a mobile coiled tubing truck **250** is utilized for deployment of the slickline equipped coiled tubing **215** via a conventional gooseneck injector **255** with supportive rig **245**. Thus, the coiled tubing **215** may be forcibly injected past pressure control equipment **265** and through the tortuous well **280** for a logging application. The depicted application may benefit from the slickline **200** having such electrically conductive, yet cohesive capacity. For example, as noted above, the slickline **200** is more cohesive than a void space metallic tube of fiber optics often used in coiled tubing operations but which is otherwise more prone to pinhole defects, rigidity, collapse and quicker erosion.

(20) To the contrary, as shown in FIG. 2B, and detailed further in reference to FIGS. 3A and 3B below, the core **100** is embedded in a more cohesive polymer layering. This is done without the use of significant void space and while maintaining an electrically insulated member **150** within the core **100** that does not substantially add to the overall profile of the line **200** (see FIG. 1). In fact, as detailed further below, the cohesive manner of layering may allow for the inclusion of a variety of additional features. For example, with specific reference to FIG. 2B, the slickline **200** in the well **280** may be exposed to hydrogen **250**. However, a cohesively layered line **200** provides a platform for the addition of features (**501**) that may serve as safeguards against hydrogen attack on the

optical fibers **120**, **125**, which may cause a darkening effect on the fibers **120**, **125** and/or thermal expansion (see FIGS. 5A and 5B). Thus, telemetric fiber optic capacity of the line **200** may be better protected overall.

(21) As depicted in FIG. 2B, the slickline **200** includes a core **100** that is not limited solely to fiber optics floating within a metal tube, but is rather of a more cohesive unitary form (e.g. fiber optics **120**, **125** retained by filler **140** in one embodiment described above). Thus, as indicated, conveyance via coiled tubing **215** with such an electrically conductive fiber optic slickline **200** disposed in the interior flow passage **217** thereof is less likely to subject the line **200** to pinhole defects, collapse and other failure modes during the coiled tubing application.

(22) Referring now to FIGS. 3A and 3B, cross sectional views of the fiber optic electrical core **100** of FIG. 1 are depicted as it is assembled into the completed fiber optic slickline **200** of FIGS. 2A and 2B. Specifically, FIG. 3A depicts the core **100** with different layers **300**, **325** provided there-over, whereas FIG. 3B depicts the completed slickline **200** complete with outer cladding **375**.

(23) With specific reference to FIG. 3A, the core **100** may be surrounded by a synthetic fiber layer **300**. For example, in one embodiment, this layer is a para-aramid synthetic fiber such as Kevlar® for sake of enhancing overall strength. However, carbon, epoxy glass, an aromatic polyester such as Vectran®, or a variety of other material types may be utilized depending on desired characteristics of the completed line **200**.

(24) With added reference to FIG. 3B, a comparatively thinner and lighter weight polymer layer **325** may be extruded over the underlying fiber layer **300** to serve as an adherent. That is, adherence between a subsequent cladding layer **375** and the fiber layer **300** may be enhanced by way of the intervening adherent layer **325**. In the embodiment shown, the cladding layer **375** may be a conventional metal-based layer such as a steel jacket. In an embodiment, an austenitic nickel-chromium-based superalloy may be utilized such as Inconel®. Once more, the layering process may continue by way of additional cladding layers (i.e. double-cladding). This may include intervening polymer layers between cladding layers for added durability and flexibility. Regardless, the completed fiber optic slickline **200** is now of substantial crush and torque resistance in addition to maintaining a low profile or diameter and having power transmitting capacity.

(25) Referring now to FIG. 4, a cross-sectional view of an embodiment of a fiber optic slickline **200** is shown. In this embodiment, the same configuration of fiber optic electrical core **100** is utilized as that depicted in FIG. 1. However, in contrast to the type of layering depicted in FIGS. 3A and 3B, a foamed polymer **400** is utilized between the core **100** and the outer cladding layer **375**. In one embodiment, a fluorinated ethylene propylene may be used. The use of a foamed polymer **400** in this manner may protect the underlying core **100** during manufacturing. For example, the foamed polymer **400** may compensate for thermal expansion differences during high temperature cladding (e.g. between the copper member **150** of the core **100** and the steel-based cladding layer **375**). Similarly, the foamed polymer **400** may also absorb compressive forces during the cladding process, thereby further protecting the underlying core **100**.

(26) In an embodiment where protecting the underlying core **100** during processing is of foremost concern, a foamed polymer **400** may be utilized as detailed above, for example, as opposed to a synthetic fiber layer **300** as shown in FIGS. 3A and 3B. Of course, in other embodiments, the foamed polymer **400** may be used as well as a synthetic fiber layer **300**. For example, in an embodiment, a strength enhancing Kevlar® layer **300** may be placed about the core **100** with a foamed polymer **400** thereover. In such circumstances, the degree to which either layer **300**, **400** is reduced in thickness versus accepting a greater overall diameter of the slickline **200** may be a matter of preference.

(27) Referring now to FIGS. 5A and 5B, an embodiment of slickline **200** is depicted. In this embodiment, the underlying core **500** is of a configuration different from that depicted in FIGS. 1, 3A, 3B, and 4. Specifically, a single split shell conductive member **150** about a variety of bundled fiber optic threads **120**, **125** is not utilized. Rather each thread **120**, **125** is provided with its own

independent conductive member **550**, **551**, **552** thereabout. Specifically, a thread **120**, **125** may be surrounded by a polymeric buffer **127** as with the embodiment of FIG. **1**. However, an individual insulating layer **580** may be provided about each buffer **127** along with each half **555**, **557** of each member **550**, **551**, **552**. Accordingly, each conductive member **550**, **551**, **552** may then be surrounded by an electrically insulating polymer **575**.

(28) Continuing with reference to FIG. **5A**, each individual conductive member **550**, **551**, **552** and underlying associated thread **120**, **125** group is held in place by a filler matrix **140** similar to the embodiment of FIG. **1**. However, in the embodiment of FIG. **5A**, the core **500** also includes glass yarn members **501**. Specifically, in a triad embodiment as shown, four yarn members **501** are shown (an extra for the center of the core **500**). Use of glass yarn members **501** in this manner may provide both manufacturing and application advantages. For example, added stability may be provided during assembly. This may include structural support, during cladding of the polymer layer **180** for example. Additionally, these members **501** may help compensate for thermal expansion during processing in light of the different neighboring material types. Indeed, the members **501** may be helically wound to further enhance such a thermal compensating role.

(29) Application advantages are also provided as noted above. Specifically, glass yarn includes the inherent characteristic of hydrogen absorption. Thus, with added reference to FIG. **2B**, any influx of hydrogen **250** may be largely absorbed by the members **501** so as to lessen a potentially darkening effect on the fiber optic threads **120**, **125** during use of the slickline **200** in the well **280**.

(30) With specific reference now to FIG. **5B**, is a cross-sectional view of the completed fiber optic slickline **200** utilizing the fiber optic electrical core **500** of FIG. **5A** is shown. In this embodiment, the insulating polymer layer **180** is surrounded by a synthetic fiber layer **300**, an adherent layer **325**, and outer cladding **375** as detailed hereinabove. Of course, in other embodiments different types of layers or additional layers may also be utilized including foamed polymer and/or double cladding configurations.

(31) Referring now to FIG. **6A**, a cross-sectional view of an embodiment of an electrically conductive fiber optic slickline **200** is shown. In this embodiment, the fiber optics of the core **600** are again surrounded by an electrically conductive member **650** for sake of power transmission. However, in this embodiment, the electrically conductive member is made up of a plurality of individual conductive wires **655**, **657** as opposed to more unitary joining shell halves as depicted in other embodiments detailed hereinabove. This may be advantageous from a manufacturability and/or available materials standpoint. Further, the wires **655**, **657** may be helically wound about the optical fibers or threads **120**, a welded metal tube **680** and an insulating polymer layer **685** so as to serve a thermally compensating role as noted hereinabove (see glass yarn members **501** of FIGS. **5A** and **5B**).

(32) Regardless, the wires **655**, **657** are again imbedded within or surrounded by an electrically insulating polymer **575** with another insulating polymer layer **685** there-under. Thus, in the embodiment shown, where the welded metal tube **680** with void space **640** is utilized to accommodate free-floating fiber optic threads **120** and buffer **127**, sufficient electrical insulation is available to the member **650**. Of course, in other embodiments, the threads **120** may be disposed in a filler and/or a variety of other layering may be utilized in addition to the outer cladding **375** shown. In an embodiment, the tube **680** may be eliminated and a filler matrix **140** inserted into the space occupied by the void **640** such that the slickline **200** resembles an embodiment closer to that depicted in FIG. **1**.

(33) Referring now to FIG. **6B**, a cross-sectional view of an embodiment of an electrically conductive fiber optic slickline **200** is shown. In this embodiment, the core **601** is made up of separately bundled optical fibers or threads **120** and electrically conductive members **658**, **659**. That is, as opposed to positioning the members **658**, **659** circumferentially relative to the fiber optic threads **120**, they are separately located adjacent the fiber optics **120**. In the embodiment shown, the members **658**, **659** are again of wire form disposed in electrically insulating polymer **575** and

the fiber optics **120** are of individually buffered fibers located in a void space **640** of a metal tube **680**. As a matter of enhancing strength and thermal compensation, each of the members **658**, **659**, and the tube **680** may be helically wound relative one another similar to other embodiments detailed hereinabove.

(34) With further reference to other embodiments discussed above, a filler matrix **140** is utilized to provide a more cohesive form to the overall core **601**. Indeed, in the embodiment shown, glass yarn members **501** are again also utilized within the matrix **140** as a manufacturing and/or application aid as also detailed hereinabove. Further, an adherent layer **325** and outer cladding **375** are shown. However, once again, a variety of other layering may be utilized.

(35) Embodiments of fiber optic slickline are detailed hereinabove that include the feature of downhole power transmission over an electrically conductive member. Thus, a mode of backup telemetry to the fiber optics of the line is also provided. However, this is achieved through embodiments that strategically utilize the available space of traditionally sized slickline, that is, without substantially adding to the overall outer diameter. Thus, the advantages of utilizing slickline over much larger and more cost prohibitive wireline cabling may be retained for suitable downhole applications.

(36) In operation, when an embodiment of the fiber optic slickline **200** disposed in the interior flow passage **217** of the coiled tubing **215** and disposed in a wellbore such as the wellbore **280** shown in FIG. 2A, as will be appreciated by those skilled in the art. The relatively small exterior or outer diameter of the fiber optic slickline **200** (below 0.20 inches in outer diameter) compared to the interior flow passage **217** of the coiled tubing **215** (the outer diameter of which is typically between about 2 and $\frac{1}{8}$ inches (about 53.9 mm) to about 3 and $\frac{3}{4}$ inches (about 95.3 mm)), which advantageously does not substantially reduce the amount of internal area or volume within the interior flow passage **217** for performing wellbore operations requiring fluid flow through the interior flow passage **217**. The wellbore operation may comprise a coiled tubing operation including, but not limited to, a fluid pumping operation, a fracturing operation, an acidizing operation, and/or a drilling operation. The robustness of the fiber optic slickline **200** allows the slickline **200** to provide telemetry and/or power at an extensive range of fluid flow rates during wellbore operations (such as including flow rates up to those flow rates typically seen in hydraulic fracturing operations and the like, as will be appreciated by those skilled in the art) and fluid types (including viscous fluids such as gels and the like) while maintaining its structural integrity. The fiber optic slickline **200** and/or the coiled tubing **215** may be attached to a wellbore device or tool such as a downhole coiled tubing tool **275** or the like, as shown in FIG. 2A. In a non-limiting example, the wellbore device or tool may comprise a measurement device to measure a property and generate an output and an interface device to convert the output from the measurement device to an optical or electrical signal. The property may be any property that can be measured in a borehole such as, but not limited to, pressure, temperature, distributed temperature, pH, amount of precipitate, fluid temperature, depth, chemical luminescence, gamma-ray, resistivity, salinity, fluid flow, fluid compressibility, viscosity, compression, stress, strain, tool location, tool state, tool orientation, and combinations thereof. In some embodiments, the apparatus of the invention may comprise a device to enter a predetermined branch of a multi-lateral well. In particular embodiments, the wellbore may be a multilateral well and the measured property be a tool orientation or a tool position. Types of wellbore devices may comprise a camera, a caliper, a feeler, a casing collar locator, a sensor, a temperature sensor, a chemical sensor, a pressure sensor, a proximity sensor, a resistivity sensor, an electrical sensor, an actuator, an optically activated tool, a chemical analyzer, a flow-measuring device, a valve actuator, a firing head actuator, a tool actuator, a reversing valve, a check valve, and a fluid analyzer. The wellbore device may be provided power and telemetry by the fiber optic slickline **200** or other embodiments thereof disclosed hereinabove. A variety of wellbore operations may be performed, such as matrix stimulation, fill cleanout, fracturing, scale removal, zonal isolation, perforation, downhole flow control, downhole

completion manipulation, well logging, fishing, drilling, milling, tractoring, measuring a physical property, locating a piece of equipment in the well, locating a particular feature in a wellbore, controlling a valve, and controlling a tool, as will be appreciated by those skilled in the art.

(37) The fiber optic slickline **200** disposed within the coiled tubing **215** may provide telemetry and power for transmitting signals, power, or information from the wellbore to the surface or from the surface to the wellbore for a number of downhole operations and/or tools disposed in the wellbore on the coiled tubing, as will be appreciated by those skilled in the art. The fiber optic slickline **200** disposed within the coiled tubing **215** may be configured to obtain DTS, DPS, DSS, and/or DVS measurements utilizing the fiber optic threads **120, 125** as will be appreciated by those skilled in the art. In an embodiment, the fiber optic slickline **200** may be disposed directly into the wellbore **280** without the use of coiled tubing and may be configured to obtain DTS, DPS, DSS, and/or DVS measurements utilizing the fiber optic threads **120, 125** as will be appreciated by those skilled in the art.

(38) The preceding description has been presented with reference to presently preferred embodiments. Persons skilled in the art and technology to which these embodiments pertain will appreciate that alterations and changes in the described structures and methods of operation may be practiced without meaningfully departing from the principle, and scope of these embodiments. Regardless, the foregoing description should not be read as pertaining only to the precise structures described and shown in the accompanying drawings, but rather should be read as consistent with and as support for the following claims, which are to have their fullest and fairest scope.

Claims

1. A downhole assembly for deployment in a well at an oilfield, the downhole assembly comprising: an electrically conductive fiber optic slickline for deployment in the well, the electrically conductive fiber optic slickline comprising: a fiber optic thread disposed in a filler matrix; a metal tube disposed about and encasing the fiber optic thread and the filler matrix; an electrically conductive member circumferentially surrounding the metal tube, the fiber optic thread, and the filler matrix; an electrically insulating polymer circumferentially surrounding the metal tube, the fiber optic thread, the filler matrix, and the electrically conductive member; a synthetic fiber layer circumferentially surrounding the metal tube, the fiber optic thread, the filler matrix, the electrically conductive member, and the electrically insulating polymer; an outer cladding disposed about and encasing the metal tube, the fiber optic thread, the filler matrix, the electrically conductive member, the electrically insulating polymer, and the synthetic fiber layer; and an adherent layer disposed between and contacting the synthetic fiber layer and the outer cladding to enhance adherence between the synthetic fiber layer and the outer cladding.
2. The downhole assembly of claim 1, wherein the metal tube is a steel tube.
3. The downhole assembly of claim 1, wherein the electrically conductive member comprises copper.
4. The downhole assembly of claim 1, wherein the outer cladding is an austenitic nickel-chromium-based superalloy.
5. The downhole assembly of claim 1, wherein the electrically conductive fiber optic slickline is disposed in an interior flow passage of a coiled tubing.
6. The downhole assembly of claim 5, wherein a slickline outer diameter of the electrically conductive fiber optic slickline is less than 0.2 in.
7. The downhole assembly of claim 6, wherein a coiled tubing outer diameter is between 2 and $\frac{1}{8}$ in. and 3 and $\frac{3}{4}$ in.
8. The downhole assembly of claim 1, wherein the electrically conductive fiber optic slickline is configured to provide telemetry and power for transmitting at least one of signals, power, or information from the well to a surface location or from the surface location to the well.

9. The downhole assembly of claim 1, wherein the synthetic fiber layer includes a para-aramid synthetic fiber.
10. The downhole assembly of claim 1, wherein the electrically conductive member includes a plurality of individual conductive wires.
11. The downhole assembly of claim 1, wherein the electrically conductive fiber optic slickline further comprises a plurality of glass yarn members.
12. The downhole assembly of claim 11, wherein the plurality of glass yarn members is helically wound.
13. The downhole assembly of claim 11, wherein the plurality of glass yarn members is located between individually conductive members.
14. The downhole assembly of claim 11, wherein the plurality of glass yarn members is disposed in the filler matrix.
15. An electrically conductive fiber optic slickline for deployment in a well, the electrically conductive fiber optic slickline comprising: a fiber optic thread; a metal tube disposed about and encasing the fiber optic thread; an electrically conductive member disposed circumferentially about the metal tube and the fiber optic thread; an electrically insulating polymer disposed about the metal tube, the fiber optic thread, and the electrically conductive member; a synthetic fiber layer circumferentially surrounding the metal tube, the fiber optic thread, a filler matrix, the electrically conductive member, and the electrically insulating polymer; an outer cladding disposed about and encasing the metal tube, the fiber optic thread, the electrically conductive member, and the electrically insulating polymer, and an adherent layer disposed between and contacting the synthetic fiber layer and the outer cladding to enhance adherence between the synthetic fiber layer and the outer cladding.
16. The electrically conductive fiber optic slickline of claim 15, wherein the fiber optic thread is disposed in the filler matrix.
17. The electrically conductive fiber optic slickline of claim 15, wherein the metal tube is a steel tube.
18. The electrically conductive fiber optic slickline of claim 15, wherein the electrically conductive member comprises copper.
19. The electrically conductive fiber optic slickline of claim 15, wherein the outer cladding is an austenitic nickel-chromium-based superalloy.
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